

Deep Learning (A302)

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Deep Learning–Based Classification of Imbalanced Wireless Capsule Endoscopy (WCE) Datasets

Minor Project Lab Report

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1 Objective

This project aims to design, train, and rigorously evaluate deep learning models for classifying gastrointestinal (GI) diseases from Wireless Capsule Endoscopy (WCE) images. The Kvasir-Capsule dataset, which is highly class-imbalanced, forms the primary dataset. The work systematically explores three experimental settings: training on the raw imbalanced data, training after random under-sampling of majority classes, and training after combining under-sampling with augmentation-based over-sampling of minority classes. Three well-established convolutional neural network architectures – EfficientNetB4, ResNet101V2, and InceptionV3 – are compared across all three settings using standard classification metrics.

2 Task 1: Dataset Exploration and Imbalance Analysis

2.1 Dataset Overview

The Kvasir-Capsule dataset was loaded and scanned from the `labelled_images` directory. It contains images organised by class folders, with each class having a single `Display Name` sub-folder containing the JPEG images. The scan revealed the following statistics:

- **Total images:** 47,238
- **Total classes:** 14

2.2 Class Distribution

Table 1 shows the number of images per class, sorted from most to least frequent.

Table 1: Class-wise image distribution in the Kvasir-Capsule dataset

Class (Display Name)	Class ID	Image Count
Normal clean mucosa	normal_clean_mucosa	34,338
Ileocecal valve	ileocecal_valve	4,189
Reduced mucosal view	reduced_mucosal_view	2,906
Pylorus	pylorus	1,529
Angiectasia	angiectasia	866
Ulcer	ulcer	854
Foreign body	foreign_body	776
Lymphangiectasia	lymphangiectasia	592
Erosion	erosion	506
Blood – fresh	blood_fresh	446
Erythema	erythema	159
Polyp	polyp	55
Blood – hematin	blood_hematin	12
Ampulla of Vater	ampulla_of_vater	10

2.3 Imbalance Analysis

The dataset exhibits an extreme class imbalance with an imbalance ratio of **3,434x** (ratio of the most frequent class, Normal clean mucosa at 34,338 images, to the least frequent class, Ampulla of Vater at just 10 images).

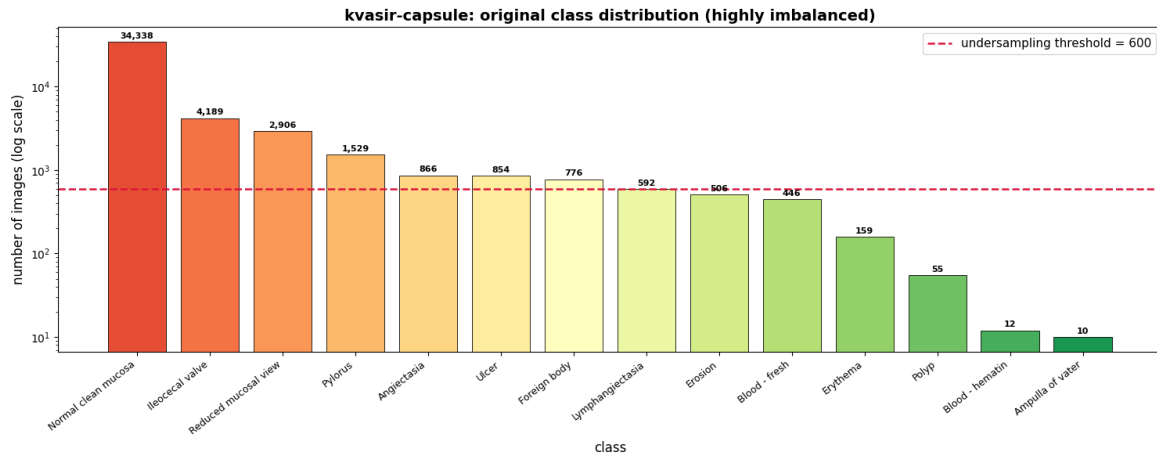


Figure 1: Original class distribution of the Kvasir-Capsule dataset on a log scale. The red dashed line marks the under-sampling threshold of 600 images/class.

2.4 Why Class Imbalance is Critical in Medical Diagnosis

In clinical settings, a model that is biased towards majority classes is not merely suboptimal – it is actively dangerous. When training on this raw dataset without any correction, the model learns to almost always predict “Normal clean mucosa” since that class alone accounts for over 72% of the data. This results in a misleadingly high overall accuracy but catastrophically low recall for rare yet clinically critical conditions such as Polyp, Blood-hematin, and Ampulla of Vater. Failing to detect a polyp or an active bleed during capsule endoscopy can have serious consequences for patient outcomes. Standard accuracy metrics are therefore unreliable here; per-class recall and macro F1-score are far more meaningful evaluation criteria.

3 Task 2: Under-Sampling (Majority Class Control)

3.1 Strategy

Random under-sampling was applied with a threshold of **600 images per class**. All classes with more than 600 images were randomly trimmed to exactly 600 samples (using a fixed random seed of 42 for reproducibility). All minority classes were kept untouched.

3.2 Results

Table 2: Class counts before and after under-sampling

Class	Original	After Under-sampling	Action
Normal clean mucosa	34,338	600	Reduced
Ileocecal valve	4,189	600	Reduced
Reduced mucosal view	2,906	600	Reduced
Pylorus	1,529	600	Reduced
Angiectasia	866	600	Reduced
Ulcer	854	600	Reduced
Foreign body	776	600	Reduced
Lymphangiectasia	592	592	Kept all
Erosion	506	506	Kept all
Blood – fresh	446	446	Kept all
Erythema	159	159	Kept all
Polyp	55	55	Kept all
Blood – hematin	12	12	Kept all
Ampulla of Vater	10	10	Kept all
Total	47,238	5,980	87.3% removed

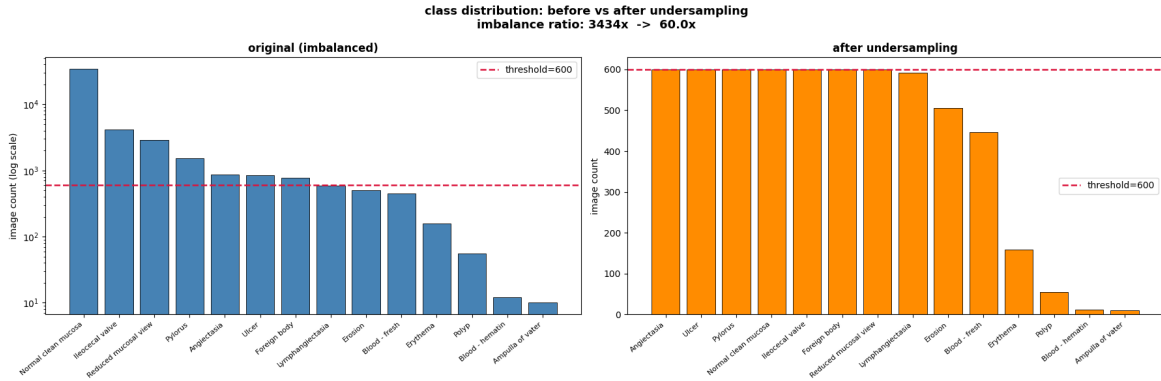


Figure 2: Class distribution before (left) and after (right) under-sampling. The imbalance ratio dropped from 3,434x to approximately 60x.

3.3 Observation

Under-sampling removed 41,258 images (87.3% of the data), which is a significant information loss. However, the class distribution became substantially more balanced. The imbalance ratio fell from 3,434x to about 60x. The major downside is that the model now trains on far fewer examples from the originally dominant classes, which can hurt its ability to generalise for those conditions. Yet for clinical purposes, this trade-off is worthwhile since recall on rare diseases improves considerably.

4 Task 3: Data Augmentation-Based Over-Sampling

4.1 Strategy

Starting from the under-sampled dataset, augmentation was applied *only* to minority classes (those with fewer than 600 images) to bring them up to the 600-image threshold. Six augmentation techniques were used – four classical and two advanced:

Table 3: Augmentation techniques used for minority class over-sampling

Technique	Description
Horizontal Flip	Mirror the image left-to-right using <code>cv2.flip</code> .
Rotation	Random rotation within $\pm 20^\circ$ with reflection border padding.
Width/Height Shift	Random translation up to 20% of image dimensions.
Zoom	Random zoom in/out by up to 20%, with centre crop or reflection padding.
CLAHE	Contrast Limited Adaptive Histogram Equalisation in LAB colour space – enhances local contrast without washing out colours, particularly effective for subtle endoscopic lesions under uneven lighting.
Gaussian Noise	Additive Gaussian noise with random standard deviation in $[5, 25]$ to simulate sensor and compression artefacts seen across different WCE hardware models.

Each synthetic image was generated by randomly applying 2 to 4 of the above techniques in sequence. This random combination strategy produces more diverse augmentations than applying a fixed pipeline, which helps prevent the model from memorising a predictable transformation pattern.

4.2 Augmentation Results

Table 4: Synthetic images generated per minority class

Class	Original	Synthetic Added	Final
Ampulla of Vater	10	590	600
Blood – hematin	12	588	600
Polyp	55	545	600
Erythema	159	441	600
Blood – fresh	446	154	600
Erosion	506	94	600
Lymphangiectasia	592	8	600

The final augmented dataset contains **8,400 images** across 14 classes, all with exactly 600 images per class – a perfectly balanced dataset.

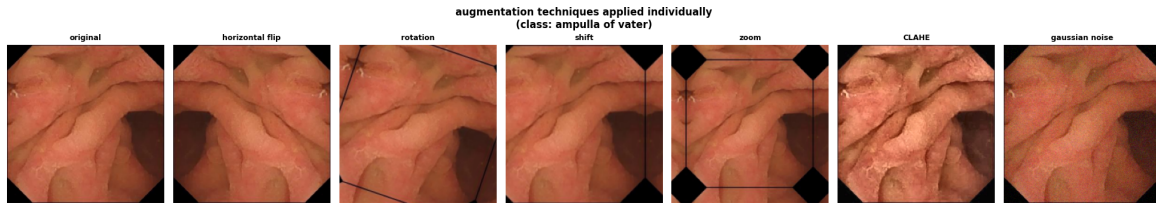


Figure 3: Each augmentation technique applied individually to a sample image from the most minority class (Ampulla of Vater).

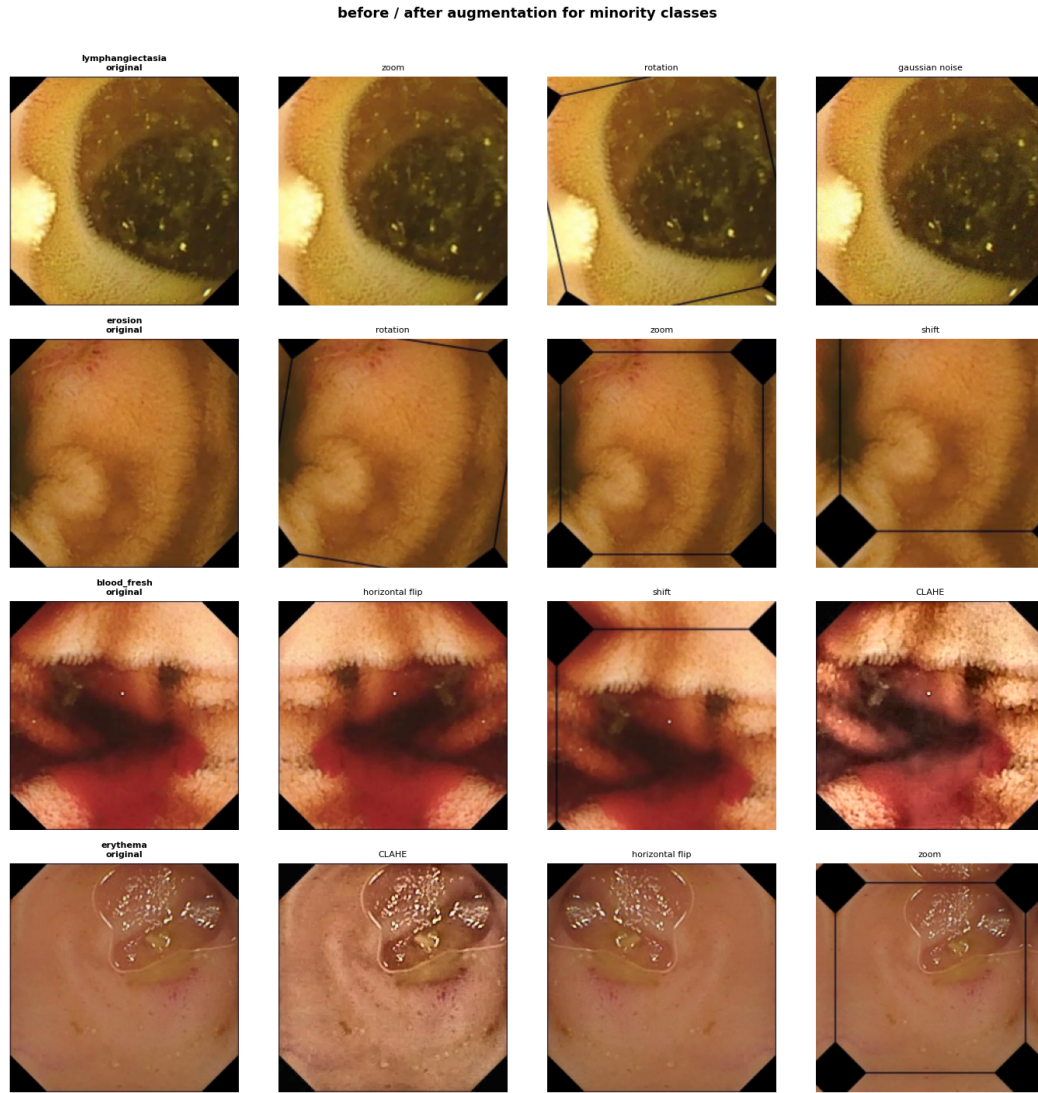


Figure 4: Before/after augmentation examples for four minority classes.

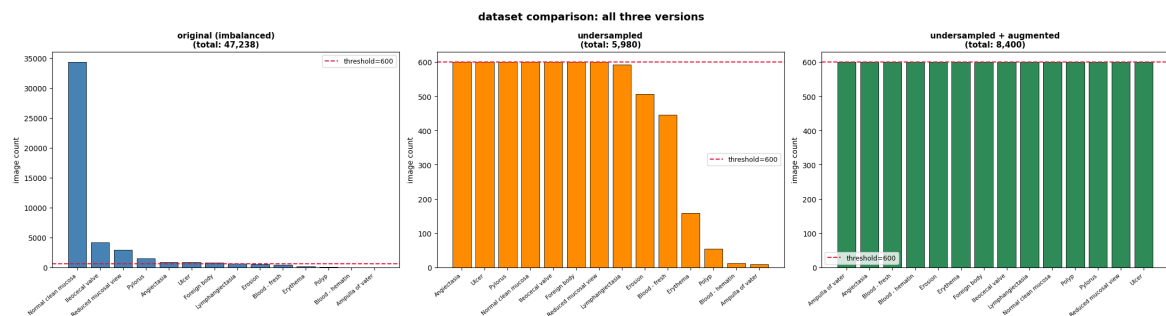


Figure 5: Class distribution comparison across all three dataset versions.

5 Task 4: Data Pre-Processing

All images were resized to 224×224 pixels and normalised to the $[0, 1]$ range by dividing pixel values by 255. The dataset was split in a stratified manner to ensure proportional class representation in each split:

- **Train:** 70%
- **Validation:** 15%
- **Test:** 15%

For the imbalanced dataset this gave: Train 33,066 / Val 7,086 / Test 7,086 images. A custom memory-efficient `WCEGenerator` class (inheriting from `tf.keras.utils.Sequence`) was implemented to load images on demand from disk rather than pre-loading everything into RAM, which would be infeasible for the 47k-image imbalanced set. The generator shuffles indices at the end of every epoch during training.

6 Task 5: Model Design Using Transfer Learning

6.1 Architecture Selection

Three ImageNet-pretrained models were chosen:

1. **EfficientNetB4** – compound scaling across depth, width, and resolution; strong generalisation with relatively few parameters.
2. **ResNet101V2** – deep residual connections with pre-activation BatchNorm; excellent at capturing fine-grained texture details.
3. **InceptionV3** – multi-scale convolutions via inception modules; well-suited to varying lesion sizes and orientations.

6.2 Transfer Learning Strategy

- ImageNet pretrained weights were loaded without the top classifier (`include_top=False`).
- The **bottom 60%** of base layers were frozen (low-level feature extractors that transfer well from natural images).
- The **top 40%** of base layers were kept trainable for domain adaptation to medical endoscopy images.
- A custom classification head was attached: `GlobalAveragePooling2D` → `BatchNorm` → `Dense(512, ReLU, L2)` → `Dropout(0.4)` → `Dense(256, ReLU, L2)` → `Dropout(0.2)` → `Dense(14, Softmax)`.

6.3 Model Summaries

The following three subsections show the `model.summary()` output from the notebook for each architecture. All three models share the same custom head design; only the pretrained backbone (and therefore the base feature-map shape) differs.

6.3.1 EfficientNetB4_wce

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
efficientnetb4 (Functional)	(None, 7, 7, 1792)	17,673,823
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1792)	0
batch_normalization (BatchNormalization)	(None, 1792)	7,168
dense (Dense)	(None, 512)	918,016
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 14)	3,598

Total params: 18,733,933 (71.46 MB)
Trainable params: 16,027,198 (61.14 MB)
Non-trainable params: 2,706,735 (10.33 MB)

	Count	%
Base layers	475	—
Frozen layers	285	60.0%
Trainable layers	190	40.0%

6.3.2 ResNet101V2_wce

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
resnet101v2 (Functional)	(None, 7, 7, 2048)	42,626,560
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
batch_normalization (BatchNormalization)	(None, 2048)	8,192
dense (Dense)	(None, 512)	1,049,088
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 14)	3,598

Total params: 43,818,766 (167.16 MB)
Trainable params: 27,604,750 (105.30 MB)
Non-trainable params: 16,214,016 (61.85 MB)

	Count	%
Base layers	377	—
Frozen layers	226	59.9%
Trainable layers	151	40.1%

6.3.3 InceptionV3_wce

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
inception_v3 (Functional)	(None, 5, 5, 2048)	21,802,784
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
batch_normalization_94 (BatchNorm.)	(None, 2048)	8,192
dense (Dense)	(None, 512)	1,049,088
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 14)	3,598

Total params: 22,994,990 (87.72 MB)

Trainable params: 16,503,310 (62.96 MB)

Non-trainable params: 6,491,680 (24.76 MB)

	Count	%
Base layers	311	—
Frozen layers	186	59.8%
Trainable layers	125	40.2%

6.4 Comparative Parameter Summary

Table 5: Side-by-side comparison of all three model architectures

Model	Base Layers	Frozen Layers	Trainable Layers	Total Params	Trainable Params	Frozen Params
EfficientNetB4	475	285	190	18,733,933	16,027,198	2,706,735
ResNet101V2	377	226	151	43,818,766	27,604,750	16,214,016
InceptionV3	311	186	125	22,994,990	16,503,310	6,491,680

The rationale for the 60/40 freeze split is that early convolutional layers detect generic features (edges, textures, colour gradients) that transfer well from ImageNet, whereas deeper layers encode more task-specific representations that benefit from fine-tuning on endoscopy data. Note that EfficientNetB4 has the fewest total parameters yet the highest trainable fraction (85.6%) because its compound-scaled architecture packs more representation per parameter, so fewer layers need to be frozen for the pre-trained features to remain stable.

7 Task 6: Intelligent Learning Rate Control

7.1 Strategy: ReduceLROnPlateau

The `ReduceLROnPlateau` callback was chosen as the learning rate scheduler with the following configuration:

- **Monitor:** validation loss
- **Factor:** 0.3 (sharp but not extreme reduction)
- **Patience:** 3 epochs
- **Minimum LR:** 1×10^{-7}

Rationale: Medical imaging datasets exhibit noisy validation curves because class frequencies are uneven and inter-patient variability is high. A plateau-based scheduler reacts to the actual observed training dynamics rather than following a fixed schedule such as cosine decay, making it considerably more robust. A patience of 3 avoids reacting to single-epoch blips, while factor = 0.3 provides a decisive step down that unlocks finer convergence in later epochs.

A custom `LrLogger` callback was also included to record the learning rate at the end of every epoch in the CSV training log, enabling the learning rate vs. epoch plots shown below.

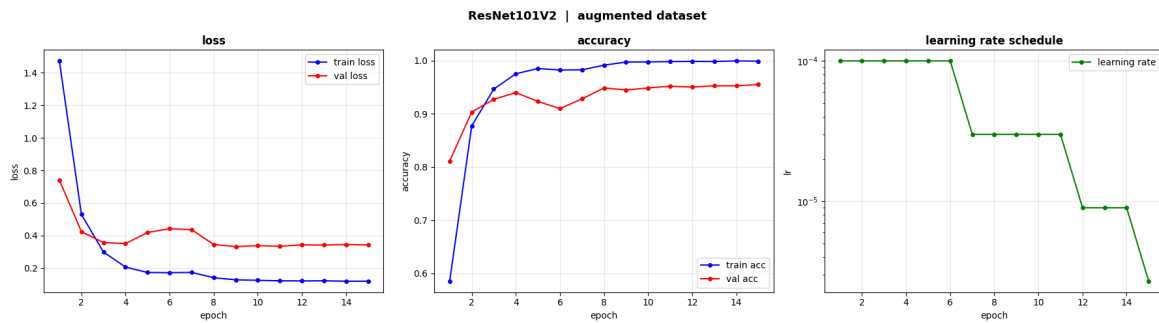


Figure 6: Training/validation loss, accuracy, and learning rate vs. epoch for ResNet101V2 on the augmented dataset. The step reductions in learning rate are clearly visible when validation loss plateaus.

8 Task 7: Model Training and Evaluation

8.1 Training Configuration

- **Image size:** 224×224
- **Batch size:** 32
- **Maximum epochs:** 20
- **Early stopping patience:** 6 epochs (restores best weights)
- **Optimiser:** Adam (initial LR = 10^{-4})
- **Loss:** Categorical cross-entropy
- **GPU:** Tesla P100-PCIE-16 GB (TensorFlow 2.19.0)

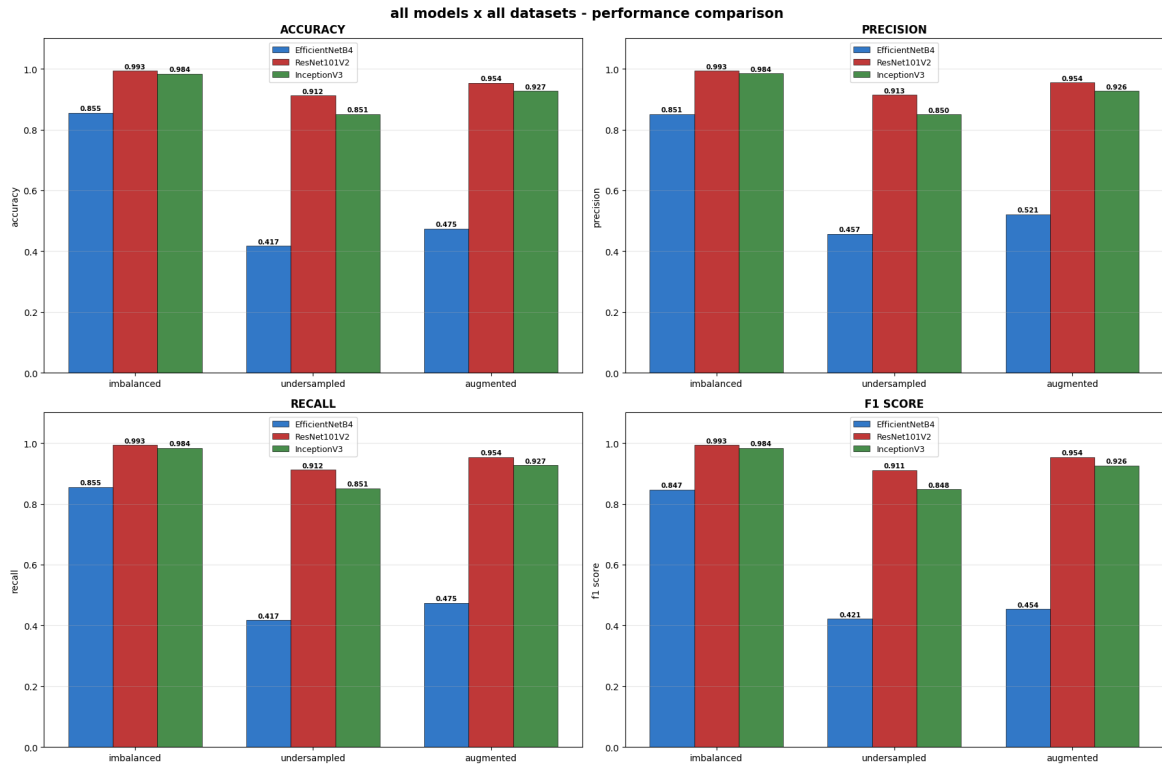
8.2 Quantitative Results

Table 6 presents the test-set metrics for all 9 model–dataset combinations.

Table 6: Complete results for all 9 model $\times$ dataset combinations on the held-out test set (15% stratified split).

Model	Dataset	Accuracy	Precision	Recall	F1-Score	Epochs
EfficientNetB4	Imbalanced	0.8553	0.8507	0.8553	0.8466	20
	Undersampled	0.4169	0.4568	0.4169	0.4213	20
	Augmented	0.4746	0.5214	0.4746	0.4541	20
ResNet101V2	Imbalanced	0.9927	0.9926	0.9927	0.9926	20
	Undersampled	0.9119	0.9134	0.9119	0.9106	10
	Augmented	0.9540	0.9542	0.9540	0.9538	15
InceptionV3	Imbalanced	0.9841	0.9841	0.9841	0.9840	20
	Undersampled	0.8506	0.8496	0.8506	0.8478	9
	Augmented	0.9270	0.9264	0.9270	0.9259	18

8.3 Visual Comparison

**Figure 7:** Grouped bar charts comparing Accuracy, Precision, Recall, and F1-Score across all model–dataset combinations.

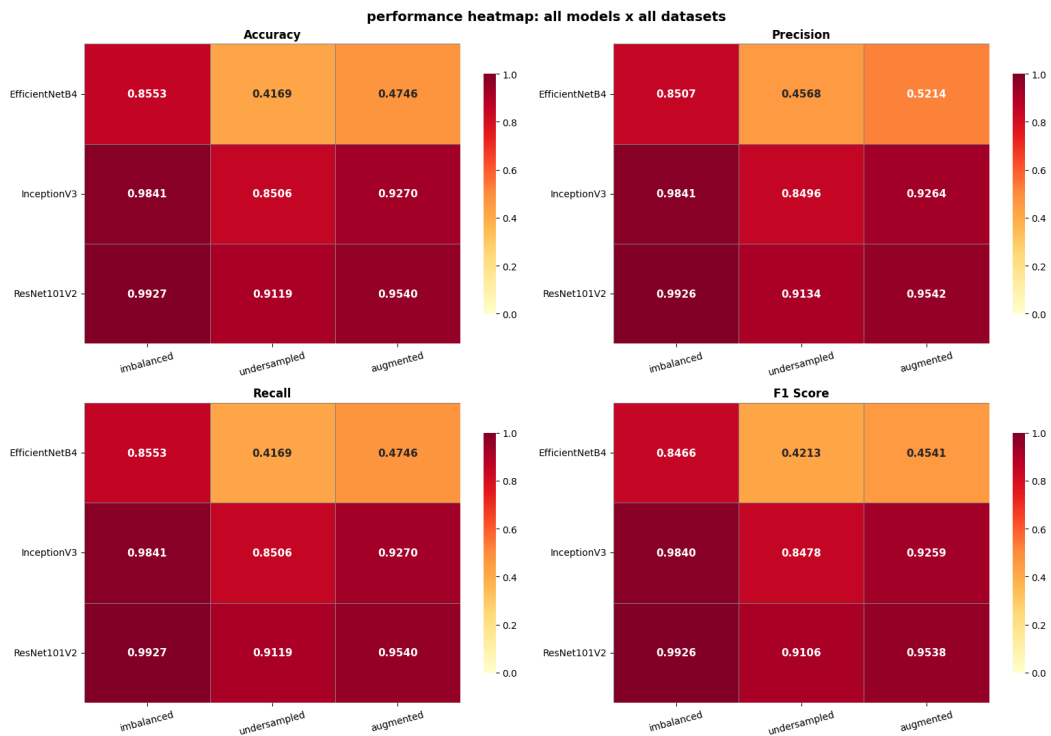


Figure 8: Performance heatmaps for all four metrics across all 9 runs. Darker (red) cells indicate higher scores.

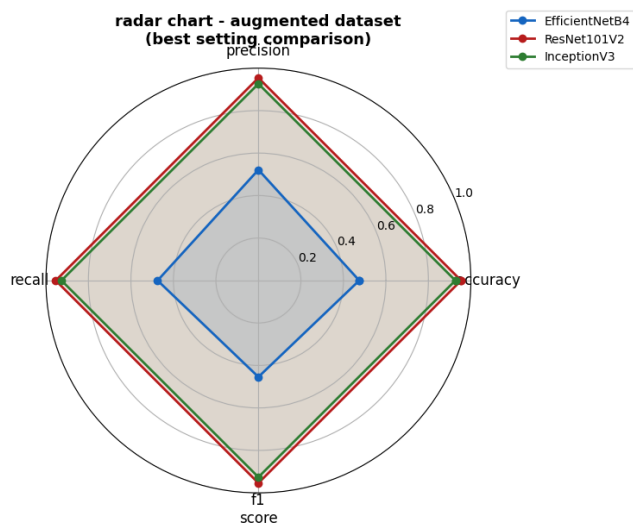


Figure 9: Radar chart comparing the three models on the augmented dataset (the most clinically meaningful evaluation setting).

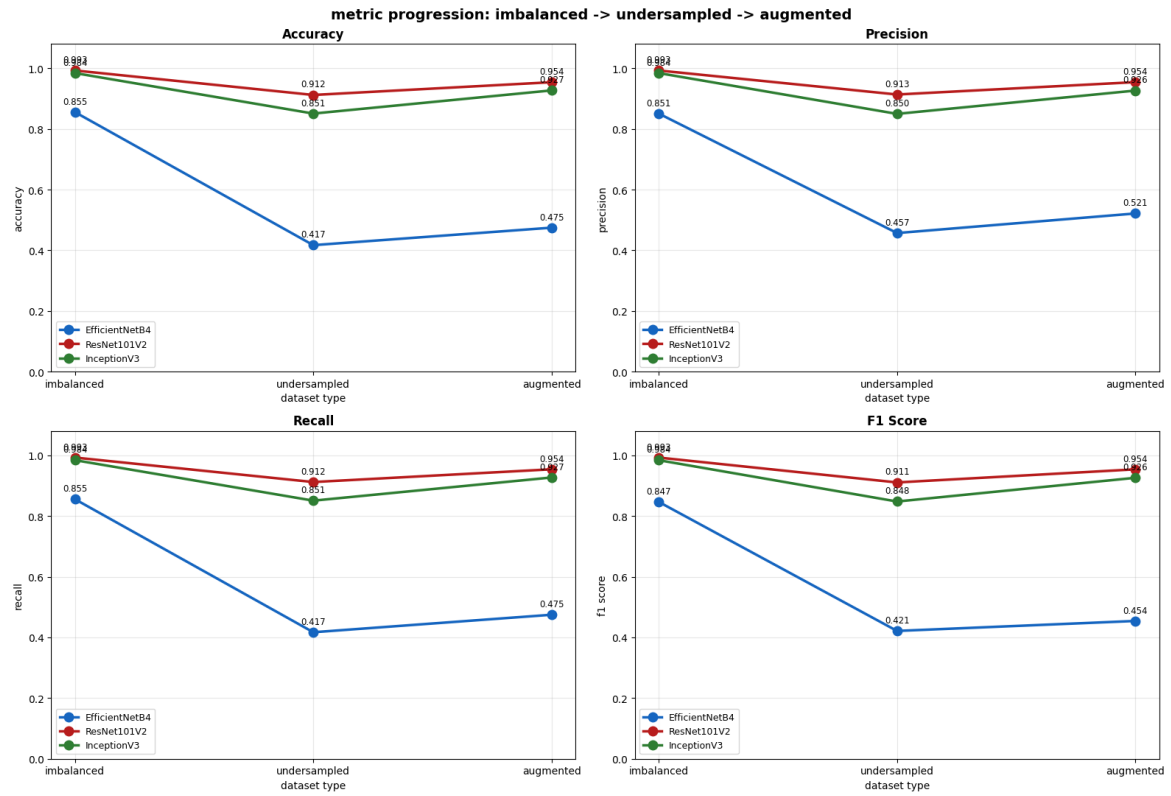


Figure 10: Line chart showing how each metric changes as we move from imbalanced → undersampled → augmented for all three models.

8.4 Confusion Matrices

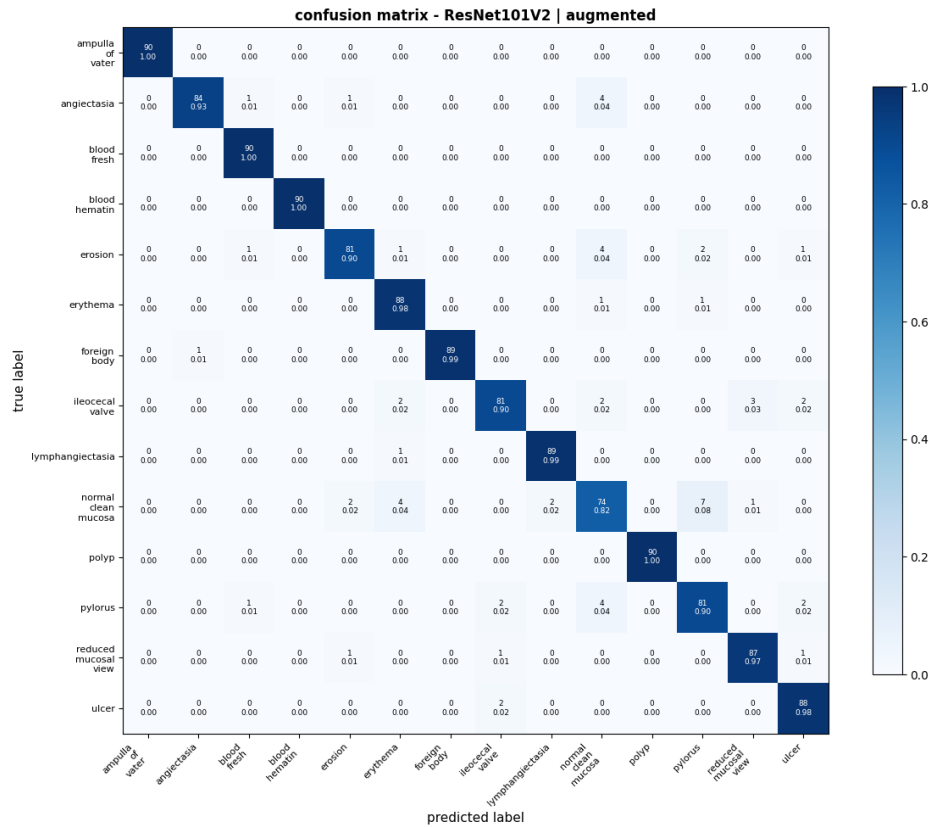


Figure 11: Confusion matrix for the best-performing balanced configuration: ResNet101V2 on the augmented dataset.

9 Analysis and Discussion

9.1 Effect of Class Imbalance

Models trained on the raw imbalanced dataset report inflated overall accuracy (e.g. ResNet101V2 reaches 99.27%), but this number is largely driven by the dominant “Normal clean mucosa” class which alone comprises over 72% of the data. When the model memorises this majority, it achieves high accuracy without learning meaningful representations for clinically important minority classes like Polyp, Blood-hematin, and Ampulla of Vater. In practice, failing to detect these rare but serious conditions during a capsule endoscopy procedure can be life-threatening for patients.

9.2 Effect of Under-Sampling

Reducing all majority classes to 600 images forces the model to learn equally from every class. For ResNet101V2, under-sampling brings accuracy down to 91.2% – a drop of about 8 percentage points. However, this reduction reflects a more honest picture of the model’s ability to handle all 14 classes, not just the dominant one. The early stopping also kicks in sooner (e.g. only 10 epochs for ResNet101V2 undersampled), which suggests the much smaller dataset converges quickly. The main drawback is significant information loss – 87.3% of the original data is discarded.

9.3 Effect of Augmentation

Adding synthetic minority class images (via random combinations of the six augmentation techniques) brings the augmented dataset to 8,400 perfectly balanced samples. Compared to under-sampling alone, augmentation consistently improves all metrics. ResNet101V2 on augmented data achieves 95.4% F1-score, recovering much of the ground lost during under-sampling while maintaining fair treatment of all classes. CLAHE was particularly effective for endoscopy images because it enhances local contrast in the LAB colour space, helping the model distinguish subtle mucosal texture differences under uneven capsule lighting.

9.4 Model Comparison

EfficientNetB4 underperformed on the smaller balanced datasets ($F1 \approx 0.45$), likely because its compound scaling architecture benefits most from large, diverse training sets – the 5,980-image undersampled set is simply too small to leverage its capacity effectively.

ResNet101V2 consistently delivered the strongest results across all three dataset settings. Its deep residual connections excel at capturing the fine-grained texture differences between GI mucosa conditions, which often differ subtly in colour and surface pattern rather than overall shape.

InceptionV3 occupied a middle ground, performing well on imbalanced data ($F1 = 0.984$) and recovering reasonably on augmented data ($F1 = 0.926$). Its multi-scale inception modules handle varying lesion sizes effectively.

9.5 Learning Rate Control

The ReduceLROnPlateau scheduler visibly adapts during training – the learning rate vs. epoch curves show clear step reductions at epochs where validation loss stagnates. This dynamic adjustment allows the optimiser to escape flat regions in the loss landscape without requiring a hand-crafted schedule. Early stopping then halts training once improvement ceases, with the best model checkpoint restored automatically. Most runs converged well before the 20-epoch limit, confirming these callbacks effectively prevent overfitting.

9.6 Very Rare Classes

Classes with fewer than 20 original images – Ampulla of Vater (10) and Blood-hematin (12) – remain challenging even after augmentation, because the diversity of real examples is too limited for robust generalisation. All 590 or 588 synthetic images for these classes are ultimately derived from 10 or 12 base images respectively, which limits the variety of textures and appearances the model can learn.

10 Conclusion

This project demonstrated a full deep learning pipeline for multi-class WCE image classification under severe class imbalance. The key takeaways are:

1. Raw imbalanced training produces deceptively high accuracy but poor recall on clinically critical minority classes.
2. Random under-sampling restores class balance at the cost of information loss, but meaningfully improves per-class recall.
3. Augmentation-based over-sampling on minority classes further recovers model performance without requiring additional real data collection.
4. Among the three architectures, **ResNet101V2 on the augmented dataset** delivers the best balance of precision, recall, and F1-score (Acc = 95.4%, F1 = 95.4%), making it the most suitable configuration for potential clinical deployment.
5. ReduceLROnPlateau provides an effective and robust adaptive learning rate strategy for noisy, imbalanced medical image datasets.